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FE 5110 Financial Engineering Project
Energy Load, Temperature Tail Behavior, and Hedging
Strategies

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Table of Detailed Workload and Original Contribution of the Project

Project Title	Energy Load, Temperature Tail Behavior, and Hedging Strategies
Name	Poreddy Saikiran Reddy
Student ID	A0297745J
Itemized workload (Approximation)	
Data collection	15 hours
Literature review	10 hours
Ideation and Planning	25 hours
Programming and Improvements	35 hours
Results Interpretation	10 hours
Project Writing	15 hours
Total Workload	110 hours
Original Contribution	This project has the following original contributions: 1. Integrated Econometric–Machine Learning Framework: The study combines classical time-series modeling with modern nonlinear methods to jointly analyze load forecasting, tail dependence, and hedging effectiveness, rarely seen in energy risk literature. 2. Empirical Link Between Forecast Errors and Temperature Extremes: Explicitly quantifies how systematic forecast underprediction creates a positive correlation between load errors and degree days, forming the statistical foundation for practical weather-derivative hedging. 3. Joint Tail Modeling with EVT–Copula Approach: Applies Extreme Value Theory in combination with a Student-t copula to simulate joint temperature load extremes and compute a <i>exceedance multiplier</i>, offering a novel quantitative measure of how strongly load and temperature co-move in the tails. 4. Seasonal Asymmetry Analysis (HDD vs CDD): Introduces a comparative seasonal perspective — showing that winter (HDD) extremes exhibit higher joint coupling with load than summer (CDD) — guiding which seasons require joint hedging. 5. Application to Major U.S. Cities: Using real multi-decade data for major cities like Boston, New York, and Chicago, the study provides empirical evidence of geographical variation in temperature load dependence and hedge effectiveness, offering region-specific insights.

Energy Load, Temperature Tail Behavior, and Hedging Strategies

Abstract

This study is conducted to develop a complete framework for energy load forecasting, extreme event correlation and risk modeling, and hedging strategy across Major U.S. cities. First, using temperature and actual energy load, we model them with a Seasonal ARMA model to capture mean dynamics and go on to show clustering effect in the residuals, and subsequently apply a Seasonal GARCH model to capture time-varying variance. The residuals of Energy Load and Temperature are individually and jointly analyzed across regions to show extreme tail co-movements and quantify tail behavior using Extreme Value Theory (EVT). Subsequently, we train an XGBoost model to predict day-ahead energy load using synthetic seasonal forecast temperature, CDD, HDD, and ETF returns. This systematic underprediction bias creates a positive correlation between forecast errors and actual degree days, enabling effective hedging. Then, a hedging strategy is formulated using a linear regression for out-of-sample data, showing hedging temperature derivatives to hedge energy loads.

Key Results:

1. Residuals from Seasonal ARMA exhibit residual clustering, requiring Seasonal ARMA + GARCH modeling.
2. ARMA-GARCH captures clustering with persistence ($\alpha + \beta \approx 0.99$).
3. Heavy-tailed distributions across all cities ($\xi = 0.18-0.29$) indicate higher extreme event probability than a normal distribution for temperature.
4. Joint Probabilities of extremes for HDD are more prevalent than CDD.
5. Geographic clusters show stronger tail dependence than southern regions
6. Moderate to high Tail Coefficient Dependence of Energy Load and Temperature between the cities, confirming that weather derivatives effectively hedge extreme temperature-driven forecast errors.
7. Temperature variables (CDD/HDD) account for 50-65% of predictive power
8. HDD hedges are highly significant: Boston ($p=0.000003$), NYC ($p=0.0007$)
9. CDD hedges vary regionally: Boston ($p=0.007$), NYC ($p=0.052$ - marginal)
10. Hedge multipliers of 0.126-0.461 MW per dollar provide actionable sizing.
11. Boston temperature contracts explain 13.1% error variance, with good economic value, while its 6.9% for NYC, with minimal economic benefit.

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Introduction

With the rapid growth of artificial intelligence, electrification, and digital infrastructure, global electricity demand has surged to unprecedented levels. Meeting this demand poses major challenges — from grid modernization and renewable integration to managing the environmental impact of traditional energy sources. As a result, efficient forecasting and risk management in the energy market have become increasingly critical. Global energy consumption is projected to rise by approximately 25% by 2030 and nearly 80% by 2050, underscoring the urgency of developing more accurate predictive and hedging frameworks.

Given this accelerating demand and the volatility introduced by extreme weather events, market participants must not only forecast energy load effectively but also hedge against climate-driven risks. While conventional energy futures and options remain established instruments, weather derivatives have emerged as a valuable complement, allowing firms to manage exposure to temperature and seasonal fluctuations.

This study develops a complete framework that connects energy load forecasting, extreme event modeling, and hedging strategy design. By combining econometric models and modern machine learning methods, it demonstrates how temperature derivatives can be used to reduce forecast uncertainty and manage tail risks in the electricity market.

Literature Review

Several studies have explored how temperature patterns influence energy demand, showing that electricity load is highly sensitive to both cold and heat extremes. Traditional models like ARIMA and ARMAX have been widely used to forecast energy load because they capture seasonality and autocorrelation well (Engle, 1982; Bollerslev, 1986). However, these models assume stable variance and normal residuals, which tend to break down during extreme weather conditions. To address this, researchers have applied GARCH-type models to capture volatility clustering and time-varying uncertainty in energy markets, improving the accuracy of short-term forecasts.

When it comes to understanding rare or extreme events, Extreme Value Theory (EVT) has been shown to provide a more realistic picture of tail risks. Studies such as McNeil and Frey (2000) highlight that energy and temperature data often follow heavy-tailed distributions, meaning that extreme outcomes occur more frequently than standard models predict. In addition, copula-based models (Ferreira, 2013) are increasingly used to measure how variables behave together during extremes, going beyond simple linear correlation and helping quantify joint risks between temperature and load.

On the risk management side, weather derivatives have become a key financial tool for hedging against temperature-driven volatility. Research by Alexandridis and Zapraniš (2013) and Bemš and Aydin (2022) shows that HDD and CDD contracts can effectively offset revenue fluctuations caused by extreme weather. More recent work has started combining these approaches with machine learning models like XGBoost and neural networks (Shivarova, 2021) to improve predictive accuracy and better estimate the impact of extreme temperature events. Together, these studies form the foundation for the approach in this project — combining classical econometric models with modern machine learning to capture extremes more accurately and design more effective hedging strategies.

Temperature Derivatives

The increasing frequency and intensity of extreme weather events pose a substantial threat to global economic stability and growth. The extreme weather events have disrupted various key sectors, including energy, manufacturing, and agriculture, where yields are greatly affected by these extreme events. It's estimated that up to 30% of global GDP is directly affected by these extreme weather events, with substantial economic damage in the case events take place.

Weather derivatives are a financial product that companies can leverage in order to hedge their losses against such extreme weather events. Provided by the Chicago Mercantile Exchange (CME), currently, Futures/Options contracts are available for 18 cities across the world. With 13 cities in the US, 4 in Europe, and 1 in Asia. The weather derivatives market is seeing a rapid expansion with a 260% surge in average trading volumes in CME, rising by 260% and it's estimated that the total market is around \$25 billion. A key distinction between Insurance and Weather Derivative contracts is that payout in Insurance is limited to proving a loss of assets and might be rejected in case the event is not part of the Insurance contract. However, payout of weather derivatives is unconditional.

Temperature Derivatives, part of wider weather derivatives, deal with extremes in Temperature. Temperature derivatives are offered during two periods in a year. The CME group, as the market leader, lists standardized futures and options on Heating Degree Days (HDD)/ Cooling Degree Days (CDD) indices for major U.S. cities. Each degree below 65° F (18.33 °C) is counted towards CDD and above to HDD. The payout is quoted in US Dollars per index point in Fahrenheit. The temperature considered is the average of Maximum (Block Maxima) and Minimum (Block Minima), making the value even more skewed at the extremes.

Modeling weather derivatives has its own challenges, since, unlike traditional products, there is no underlying asset, forcing practitioners to estimate the market price for the risks. Moreover, Temperature exhibits a more seasonal and autocorrelated pattern and heteroskedasticity, violating assumptions of standard

options pricing frameworks like Black Scholes. In the next section, temperature modeling is systematically broken down into continuation iterations of improvement.

$$T_{avg} = (T_{Max} + T_{min})/2$$

$$HDD_t = \max(0, 65^\circ F - T_t)$$

$$CDD_t = \max(T_t - 65^\circ F, 0)$$

Data Preparation

Data Preparation for the project involved collecting, cleaning, and transforming the raw data into a common format. Official Temperature data is sourced from Speedwell for 6 US cities (New York, Boston, Minneapolis, Houston, Dallas, and Chicago) for a period of 20 years (1st January 2005 to 31st December 2024). The data consists of temperatures taken every hour for the above mentioned time period. The average temperature is calculated by averaging the minimum and maximum temperatures on a day. The resulting data set contains t_{avg} , t_{min} and t_{max} for the above time period.

City	Mean(°F)	Std(°F)	Min(°F)	Max(°F)	Seasonal R^2
Boston	53.0	17.5	1.5	90.5	0.816
New York	57.3	17.6	9.0	91.5	0.834
Houston	71.0	13.3	20.5	93.5	0.728
Chicago	51.3	20.3	-16.5	87.0	0.812
Dallas	67.7	16.1	8.0	97.5	0.768
Minneapolis	47.3	23.0	-20.5	90.0	0.835

Table 1: City Temperature Statistics



Figure 1: Boston Temperature Statistics

For Energy Load, the following sources are used for data collection. The process involved downloading the Excel files programmatically through scripts or manually downloading the yearly data in some cases. (See appendix for the scripts used)

- Boston (NEMASSBOST): ISO-NE region, 3,283 days
- New York (NYISO): PAL Integrated zone, 7,300 days
- Chicago (MISO Central): Mid-continent ISO, 5,844 days
- Houston (ERCOT Coast): Texas grid, 4,017 days
- Dallas (ERCOT North): Texas grid, 4,017 days
- Minneapolis (MISO): Northern territory, 5,844 days



Figure 2: Boston Energy load statistics

City	Min Load (MW)	Max Load (MW)	Avg Load (MW)	Std Dev (MW)
Boston	2,500	4,500	3,200	450
New York	8,000	13,000	10,500	1,450
Chicago	25,000	40,000	32,000	4,200
Minneapolis	8,500	11,500	9,800	850

Table 2: Energy Load Statistics

Energy ETFs are the best proxies for the prices of underlying assets. ETF prices of various fuel sources are downloaded from Yahoo Finance through API using a script from 1st January 2014 to the latest. The estimated energy generation in the US for 2023 is below:

1. Natural Gas: ~43%
2. Coal: ~16%
3. Nuclear: ~18%
4. Renewables: ~21%
5. Other: ~2%

For the energy sources, the following ETFs are selected.

- United States Natural Gas Fund (UNG):
 - Heating fuel proxy.
 - Natural gas is the #1 fuel source for US electricity generation.
- Utilities Select Sector SPDR Fund (XLU):
 - Sector hedge
 - Tracks companies that generate and distribute electricity
- iShares Global Clean Energy ETF (ICLN)
 - Renewable Clean Energy Fund
 - Solar, wind, and hydro are rapidly growing electricity sources
- Global X Uranium ETF (URA):
 - Nuclear power
 - Nuclear power provides consistent baseload electricity
- United States Oil Fund (USO):
 - Fossil fuel energy
 - Used for peak generation and price signals.
- VanEck Coal ETF (KOL):
 - Traditional generation
 - Coal is still a significant fuel source, especially for baseload power

Methodology

In this section, we will discuss the models, model parameters, and selection criteria for the best model. Subsequently, we chose the best method to model the data for residuals. First, based on seasonal trends observed in Figure 1 and Figure 2, we fit a seasonal model using mean, Linear, and Sine and Cosine functions with different lags by day of the year. We try with different harmonics (3 and 6) to find the best fit. We use mean and linear trend to capture the rise in global temperature & Energy Load, and sinusoidal waves to capture the seasons of the year.

Base Model Specification

$$y_t = \mu + \beta t + \sum_{n=1}^N (A_n \sin(2\pi n t / 365.25) + B_n \cos(2\pi n t / 365.25)) + \epsilon_t$$

Components:

- Trend: μ is for mean, and Linear βt captures long-term changes.
- Harmonics: $N \in \{3, 6\}$ captures seasonal cycles. 365.25 is used to account for a leap year.
- A_n, B_n are coefficients of Sine and Cosine waves.
- Residuals ϵ_t are the remnant residuals.

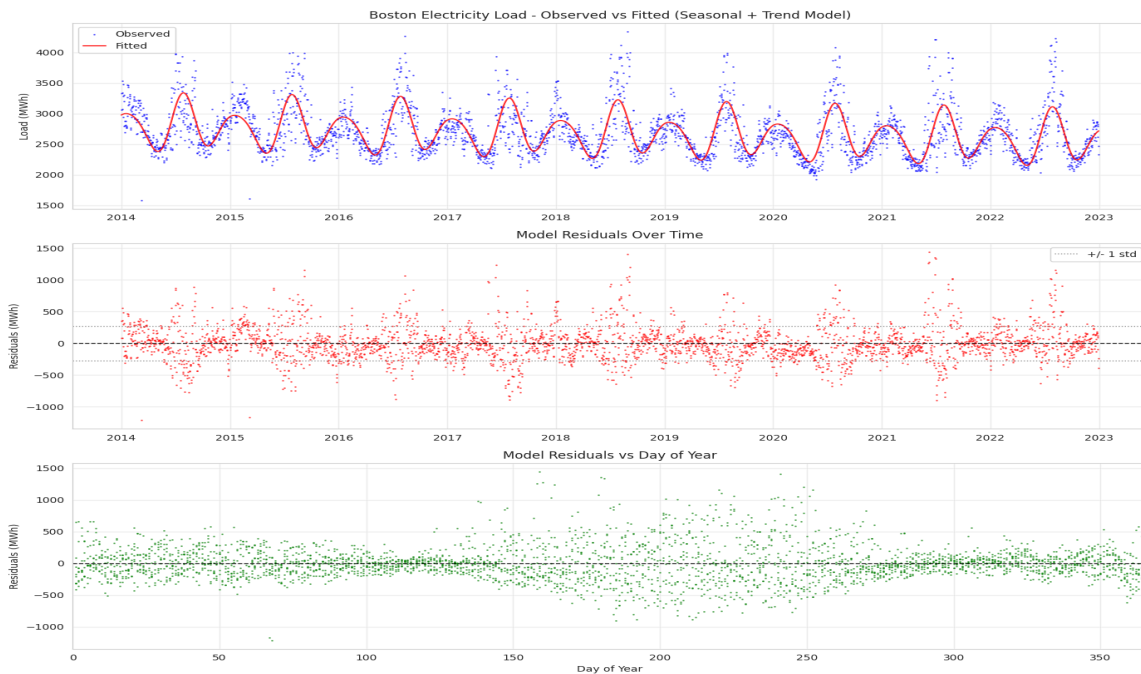


Figure 3: Boston Electricity Load Seasonal & Trend Model

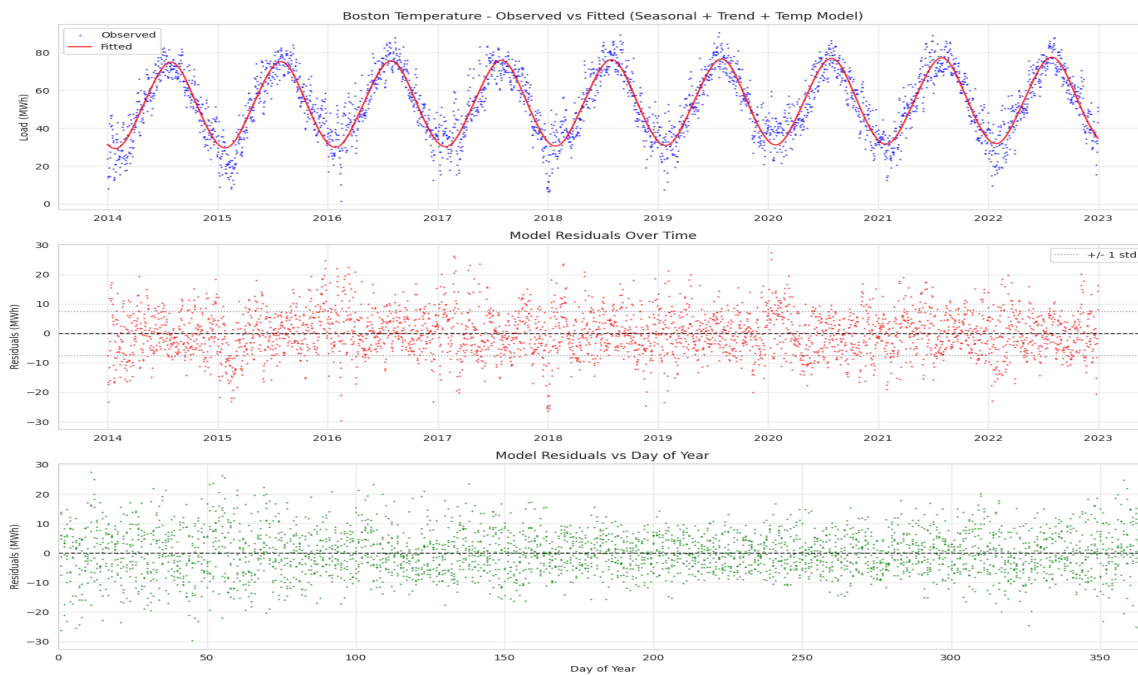


Figure 4: Boston Temperature Seasonal & Trend Model

The plot of linear and sinusoidal waves adapts to the temperature data decently. While examining the residuals, the residuals also have a clustering effect. Examining Model Residuals vs day of the year for Energy Load, the residuals have more volatility in the Day 150 to 250 period (June to September). While the results are similar for each city, Boston is chosen as an example for the Figures and Analysis.

Residual Analysis

Plotting the residuals and conducting a normality test for Energy load and temperature, the result is rejection for normality with significant lags for ACF and PACF as well.

```
=====
Boston - RESIDUAL NORMALITY TEST
=====
D'Agostino-Pearson test: statistic=477.9605, p-value=1.6301e-104
Reject normality (p < 0.05): True
=====
```

Figure 5: Residual Normality test for Boston Energy Load

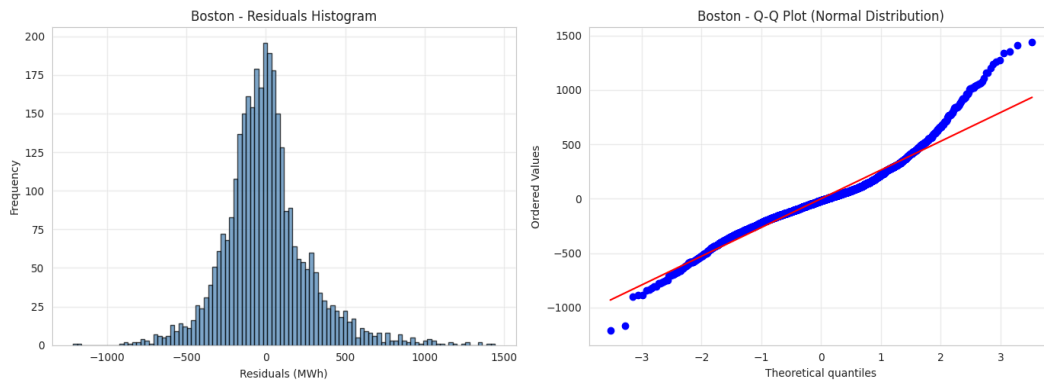


Figure 6: Residual histogram and Q-Q Plot for Boston for Energy Load

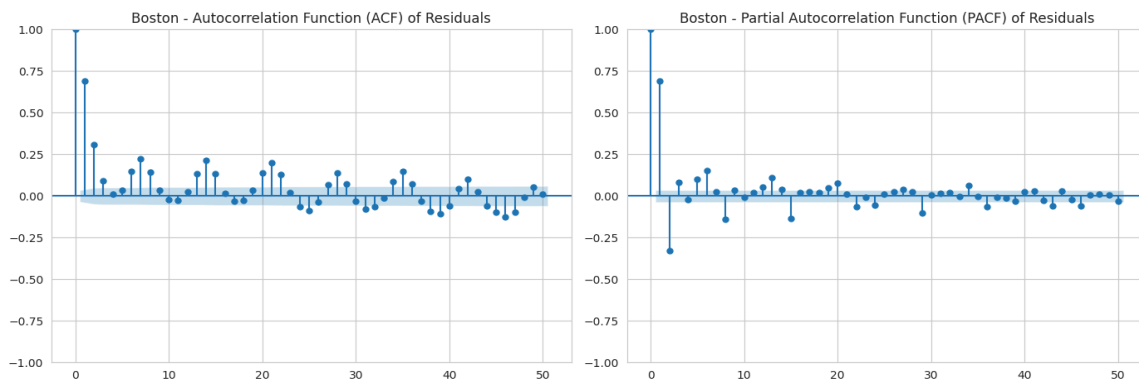


Figure 7: ACF and PACF Plot for Boston Energy Load Residuals

```
=====
Boston - RESIDUAL NORMALITY TEST
=====
D'Agostino-Pearson test: statistic=18.6081, p-value=9.1053e-05
Reject normality (p < 0.05): True
=====
```

Figure 8: Residual Normality Test for Boston Temperature Residuals

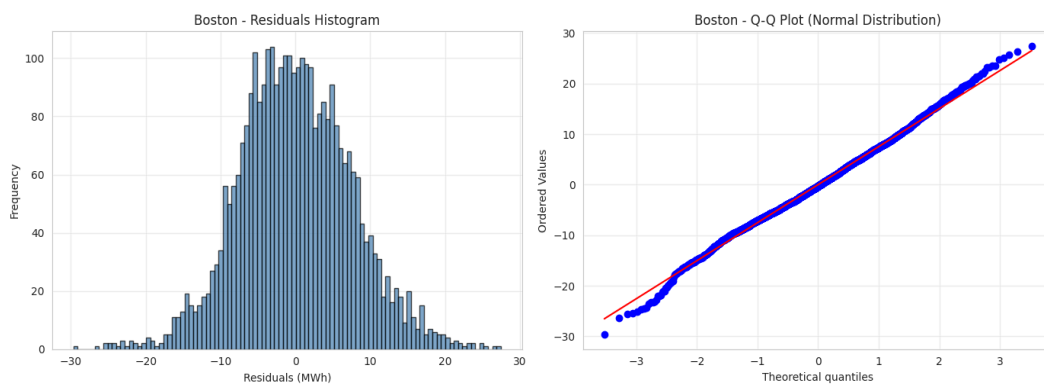


Figure 9: Residual histogram and Q-Q Plot for Boston Temperature

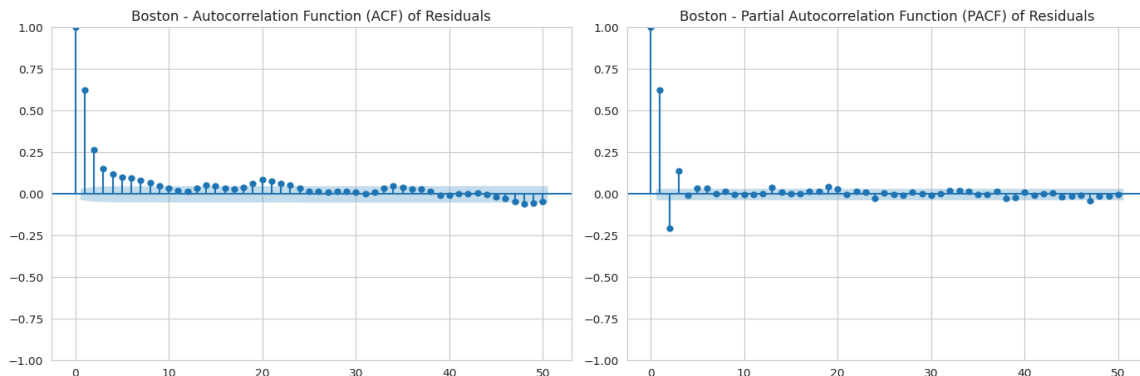


Figure 10: ACF and PACF Plot for Boston Temperature Residuals



Figure 11: Boston Energy Load Residuals plots

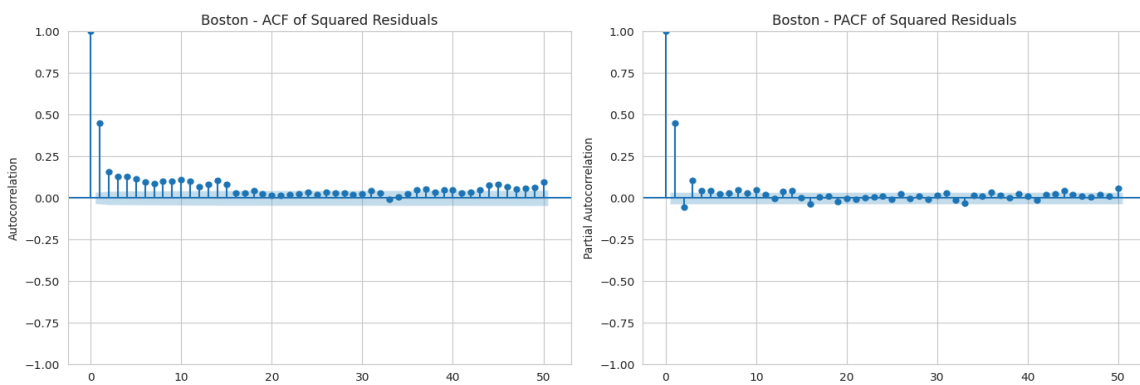


Figure 12: Boston Squared Residuals ACF and PACF

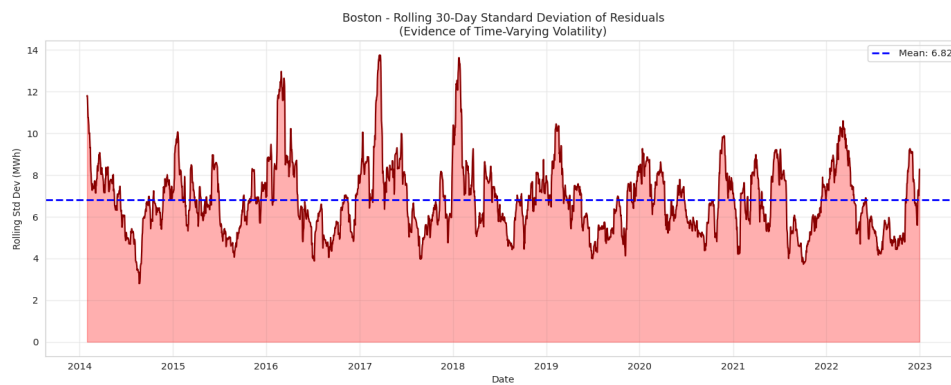


Figure 13: Boston Energy Load Rolling 30-day Standard Deviation of Residuals



Figure 14: Boston Energy Load with Trend Seasonal Model + GARCH(1,1) on Residual



Figure 15: Boston Temperature with Trend Seasonal Model + GARCH(1,1) on Residual

Diagnostic checks, including Ljung–Box and ARCH tests, confirmed that residuals exhibited serial correlation and conditional heteroskedasticity, validating the need for ARMA–GARCH modeling. Based on the above conclusion, we employ Seasonal (3,6 harmonics) + ARMA model (0,1,2,3 AR Order and 0,1,2,3 MA Order) to find the best fit for each city and consequently apply GARCH modeling to fit the residuals. The following table presents the optimal statistical models for electricity load and temperature forecasting across all regions. Models were selected based on Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), which balance goodness-of-fit with model complexity.

Model Selection

Region	Best Model	AIC	BIC	R^2	N	Persistence	Avg Volatility
Boston	3H + ARMA(1,1) + GARCH(1,2)	42,948	43,039	0.7633	15	0.9989	163.76
Chicago	3H + ARMA(3,3) + GARCH(2,3)	60,600	60,728	0.7593	21	1.0000	2,265.30
Minneapolis	3H + ARMA(1,3) + GARCH(2,3)	54,134	54,250	0.7305	19	1.0000	825.44
New York	3H + ARMA(1,3) + GARCH(1,3)	50,155	50,265	0.8490	18	0.9994	493.78

Table 3: Selection of the Best Model for Energy Load based on Region

Region	Best Model	AIC	BIC	R^2	N	Persistence	Avg Volatility
Boston	3H + ARMA(3,0) + GARCH(1,3)	20,800	20,903	0.8922	17	0.9999	5.65
Chicago	3H + ARMA(1,2) + GARCH(1,3)	20,789	20,901	0.9153	17	0.9998	5.724
Minneapolis	3H + ARMA(1,2) + GARCH(1,3)	20,870	20,973	0.9330	17	0.9999	5.813
New York	3H + ARMA(3,0) + GARCH(1,3)	20,186	20,289	0.9082	17	1.0000	5.197
Houston	3H + ARMA(1,1) + GARCH(1,2)	18,846	18,937	0.8635	17	0.9995	4.438
Dallas	3H + ARMA(1,1) + GARCH(1,2)	19834	19925	0.8843	17	0.9994	5.036

Table 4: Selection of the Best Model for Temperature based on Region

Final Model Specification

$$y_t = \mu + \beta_t + \sum_{n=1}^N (A_n \sin(2\pi nt/365.25) + B_n \cos(2\pi nt/365.25)) + ARMA(p, q) + \varepsilon_t$$

$$ARMA(p, q) = \varphi_1 y_{t-1} + \varphi_2 y_{t-2} + \dots + \varphi_p y_{t-p} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q}$$

Where residuals ε_t are modeled with GARCH(r,s)

$$\varepsilon_t = \sigma_t z_t$$

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \alpha_2 \varepsilon_{t-2}^2 + \dots + \alpha_r \varepsilon_{t-r}^2 + \beta_1 \sigma_{t-1}^2 + \beta_2 \sigma_{t-2}^2 + \dots + \beta_s \sigma_{t-s}^2$$

Model Components:

- 3H: 3 harmonic seasonal terms (annual, semi-annual, 4-month cycles)
- ARMA(p,q): Autoregressive Moving Average for residual autocorrelation
- GARCH(r,s): Generalized Autoregressive Conditional Heteroskedasticity for volatility clustering

For load models, New York records the highest explanatory power with an R^2 of 0.849, reflecting predictable patterns in a large market. All cities exhibit near-unit-root persistence ($\alpha + \beta \approx 1.0$), meaning volatility shocks persist for weeks. Chicago shows the highest volatility (2,265 MW) due to industrial activity and exposure to extreme weather. Models with three harmonics outperform more complex ones, as simplicity helps avoid overfitting under the BIC criterion.

For Temperature Models, Minneapolis achieves the highest R^2 (0.934), consistent with its strong continental seasonality. All cities display high persistence ($\alpha + \beta \approx 1.0$), indicating long-lasting temperature shocks and very low residual volatility (4–6°F) once seasonality is removed.

Model Interpretation

R^2 values range from 0.73–0.85 for load (moderate to good fit) and 0.86–0.93 for temperature (excellent fit dominated by seasonal effects). Persistence near unity confirms that GARCH modeling is essential, as constant variance assumptions are inadequate. AIC tends to favor predictive power, while BIC emphasizes parsimony. An agreement between both implies robust model selection.

In the Northeast (Boston, New York), loads show moderate R^2 and volatility with strong persistence, while the Midwest (Chicago, Minneapolis) faces higher volatility and more complex dynamics. Coastal cities (Boston, New York, Houston) are well captured by 3-harmonic models due to maritime moderation, whereas continental cities (Chicago, Minneapolis, Dallas) are more complex, but 3H is chosen by BIC and standardization to ensure compatibility across regions for correlation/TDC analysis. Across all regions, high R^2 values (0.86–0.93) confirm that temperature remains highly predictable.

Practical Applications

The models support day-ahead load forecasting, producing point forecasts with GARCH-based confidence intervals. For temperature forecasting, the ARMA component captures short-term (3–7 day) dynamics, while the harmonic component handles seasonal trends for long-range outlooks and weather derivative pricing. In risk management, GARCH volatility enables VaR/CVaR estimation, where $VaR_{95\%} = \mu_{forecast} + 1.645 \times \sigma_{GARCH}$. High persistence ($\alpha + \beta \approx 1.0$) indicates that volatility shocks last for extended periods, underscoring the need for continuous risk monitoring.

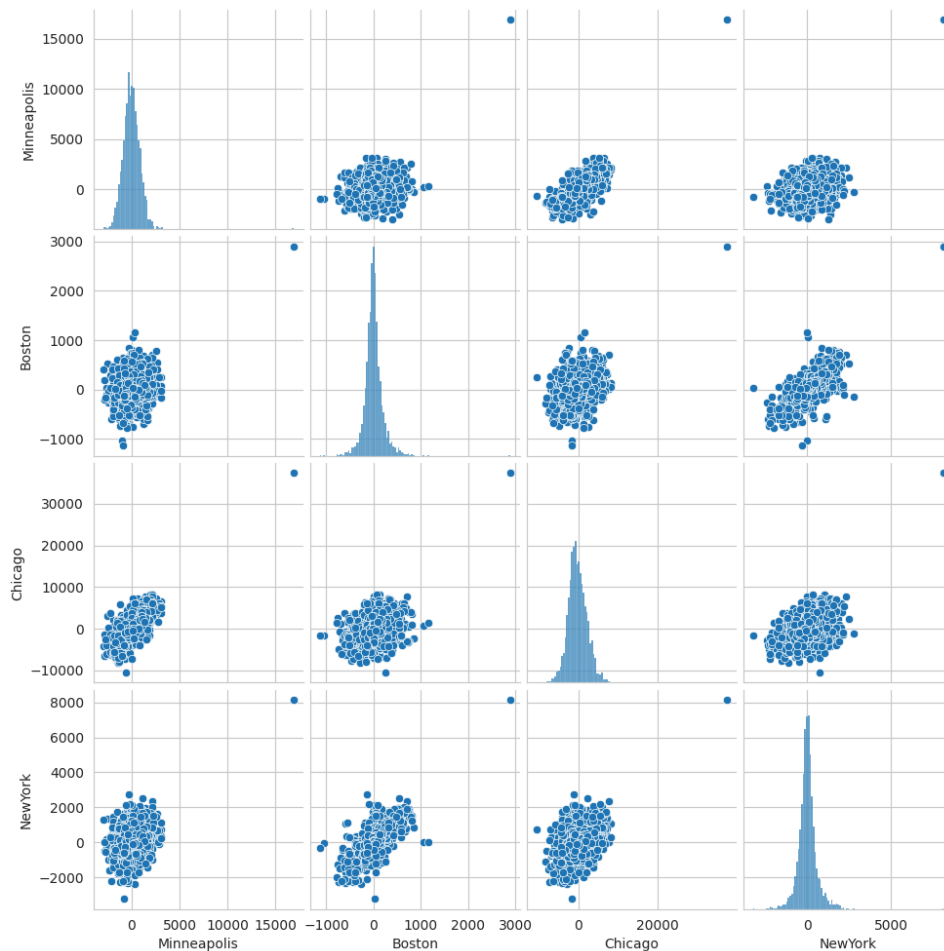


Figure 16: Pairplot of Residuals for Energy Load

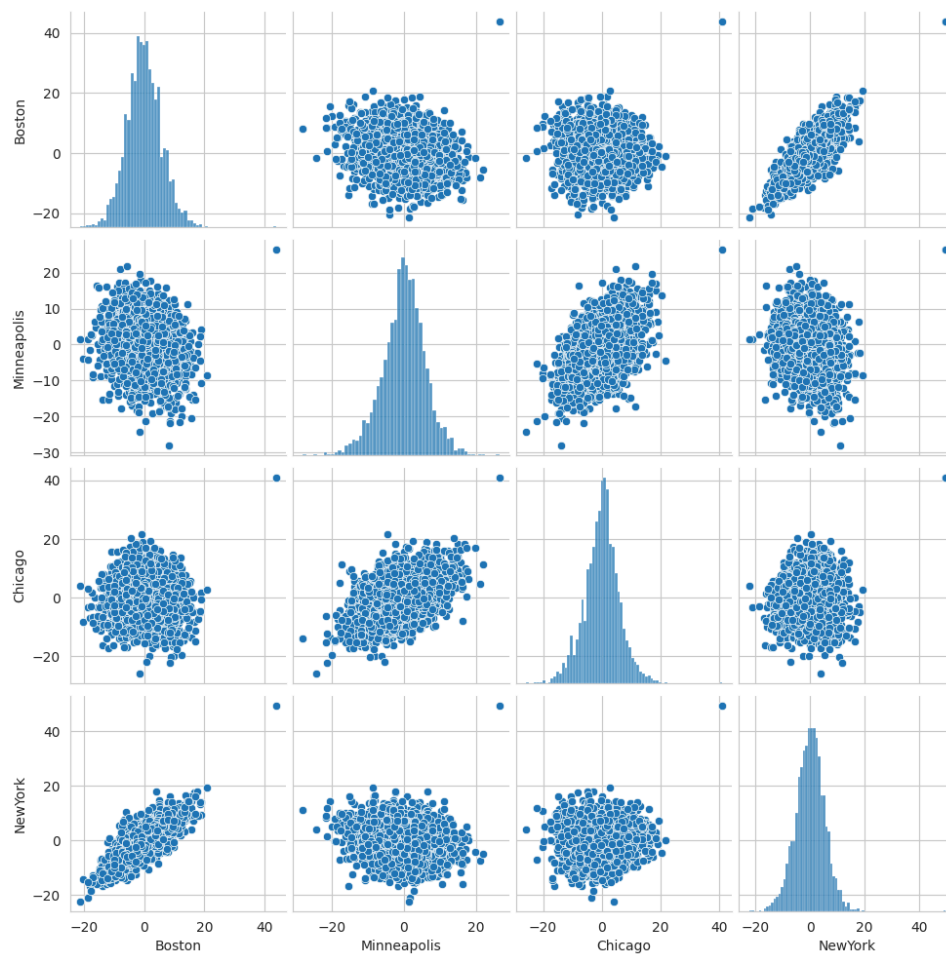


Figure 17: Pairplot of Residuals for Temperature

Correlation and Tail Dependency Coefficient

From the above pairplots for temperature and energy load residuals, a rough idea can be obtained of how the residuals move together. In order to measure how the variables move together, Kendall Tau Correlation and Ferreira Tail Dependency Coefficient are used. While the Kendall Tau measures how the variables move together by rank accordance using all the data, the values range from -1 to 1. Kendall's Tau (τ) is a rank-based correlation measure that captures monotonic dependence — i.e., how consistently two variables move in the same or opposite direction. Unlike Pearson's ρ (which looks at linear relationships), Kendall's τ focuses on the ordering of the data. Ferreira's tail dependency coefficient, on the other hand, is used to measure how strongly two random variables are linked in their extremes and uses only tail data. It's a key concept in Copula Theory and Extreme Value Analysis. The value of Ferreira TDC ranges from 0 to 1.

Definitions

Kendall's Tau

Given all pairs of observations (X_i, Y_i) and (X_j, Y_j) .

- A pair is concordant if $(X_i - X_j)(Y_i - Y_j) > 0$.
- A pair is discordant if $(X_i - X_j)(Y_i - Y_j) < 0$.

$$\tau = P(\text{concordant}) - P(\text{discordant})$$

Ferreira's Tail Dependency Coefficient

For two continuous random variables X and Y with distribution functions F_X and F_Y .

The upper tail dependence

$$\lambda_U = \lim_{u \rightarrow 1^-} P(Y > F_Y^{-1}(u) | X > F_X^{-1}(u))$$

The lower tail dependence

$$\lambda_L = \lim_{u \rightarrow 0^+} P(Y \leq F_Y^{-1}(u) | X \leq F_X^{-1}(u))$$

These capture the probability that one variable is extreme given the other is extreme, as you move towards the tails. Ferreira proposed a non-parametric estimator of the tail dependence that's more stable in finite samples.

$$\lambda = 2 - (1/(\bar{X}_k))$$

$$\lambda = 1/k(\sum_{i=1}^k (1 - \max(F_X(X_{n-i+1}), F_Y(Y_{n-i+1}))))$$

Where

- n is the sample size.
- k is the number of upper order statistics (top k observations - typically $k = n^{0.5}$ or chosen by the threshold rule.

High τ , low λ → generally move together, but not during extremes.

Low τ , high λ → usually independent, but co-crash in crises.

Results

A Lambda (λ) parameter in TDC of 0.2 to 0.5 indicates moderate tail dependence, while 0.5 to 0.8 signifies strong tail dependence. Similarly, a Kendall Tau (τ) of 0.7 to 1 indicates that variables strongly move together, while 0.3 to 0.7 indicates that they often move together.

Interpreting from the results, a moderate dependence in Kendall tau for load between New York and Boston of 0.58 and 0.62 in CDD and HDD months, respectively. For Minneapolis and Chicago, again a moderate Kendall tau of 0.5 and 0.59 in CDD and HDD months is observed, respectively. While Kendall tau shows a moderate behavior, strong tail dependence between New York and Boston of 0.68 in CDD and HDD months, and 0.6 and 0.7 between Chicago and Minneapolis in CDD and HDD months. Kendall Tau correlation matrix for both HDD and CDD periods shows a moderate correlation between Temperature and Load Residuals. The Ferreira TDC between Load and temperature residuals is around 0.4–0.5, indicating a moderate level of co-movement in the tails. This suggests that during extreme temperature conditions, when HDD/CDD payoffs are highest, the residuals tend to move in the same direction, implying the hedging strategy captures a meaningful portion of temperature-driven extreme payoffs.

While linear correlations capture average relationships, they fail to describe how variables behave jointly during extremes. Copula theory provides a flexible way to separate marginal behavior from dependence structure (Sklar's Theorem), allowing a more accurate assessment of tail co-movements, which is essential before applying EVT to quantify joint risks.

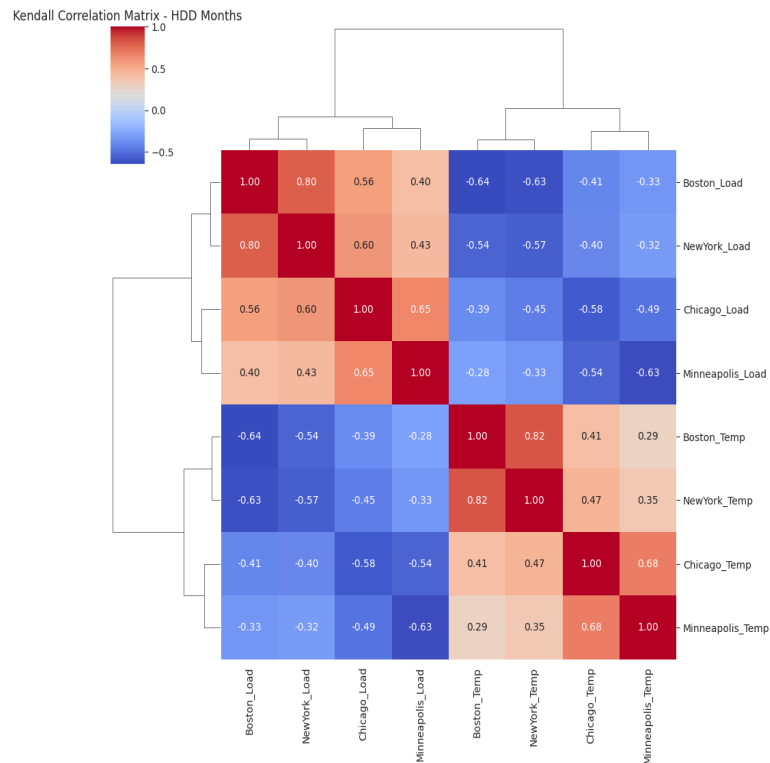


Figure 18: Kendall Tau for Load and Temperature during HDD months

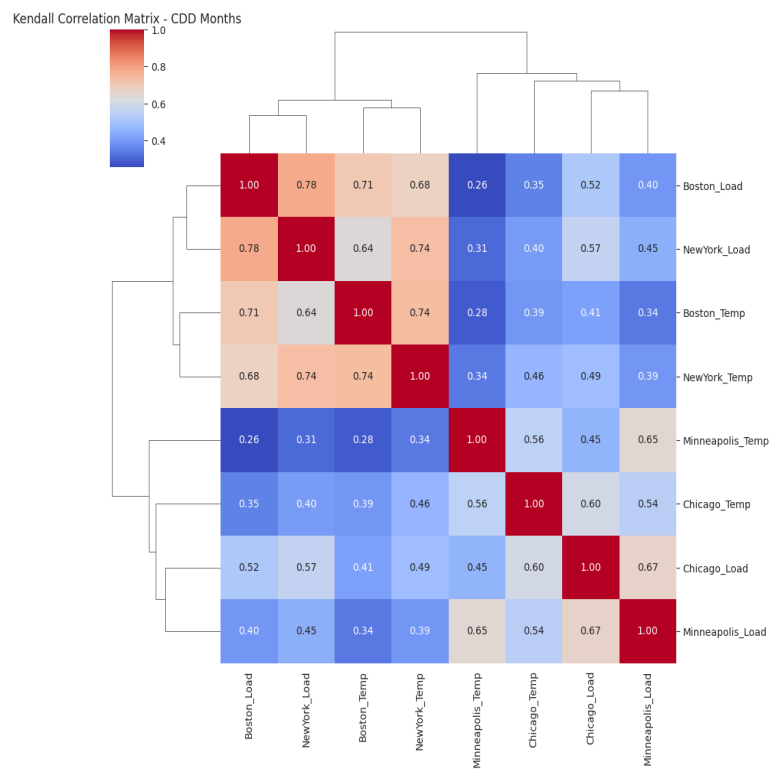


Figure 19: Kendall Tau for Load and Temperature during CDD Months

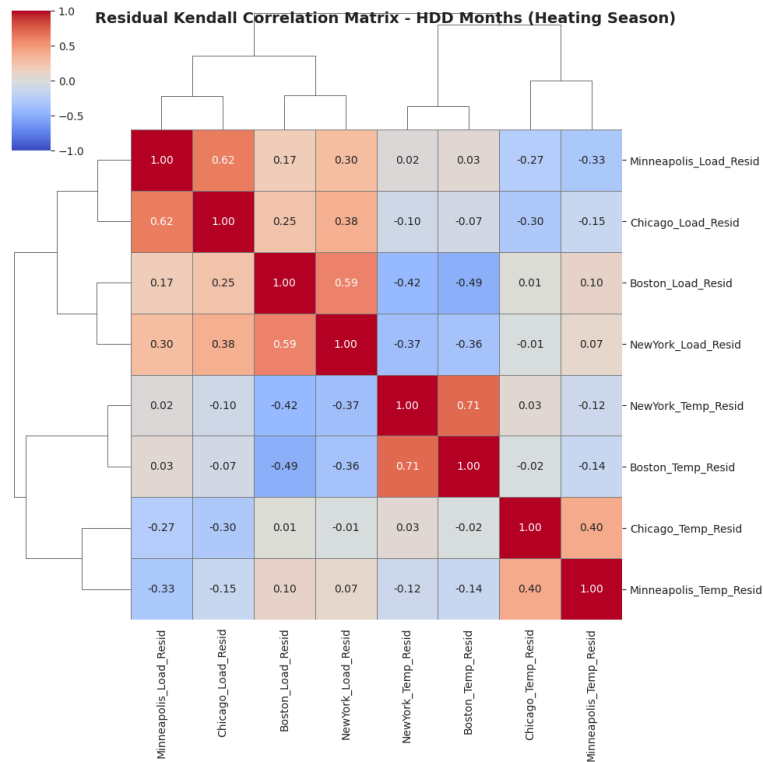


Figure 20: Kendall Tau for Load and Temperature Residuals during HDD months

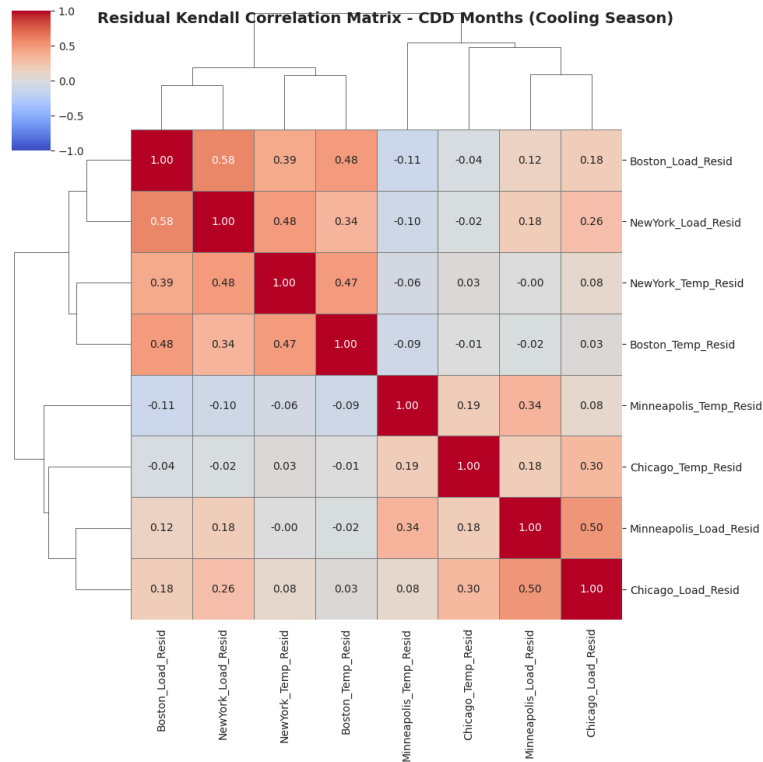


Figure 21: Kendall Tau for Load and Temperature Residuals during CDD months

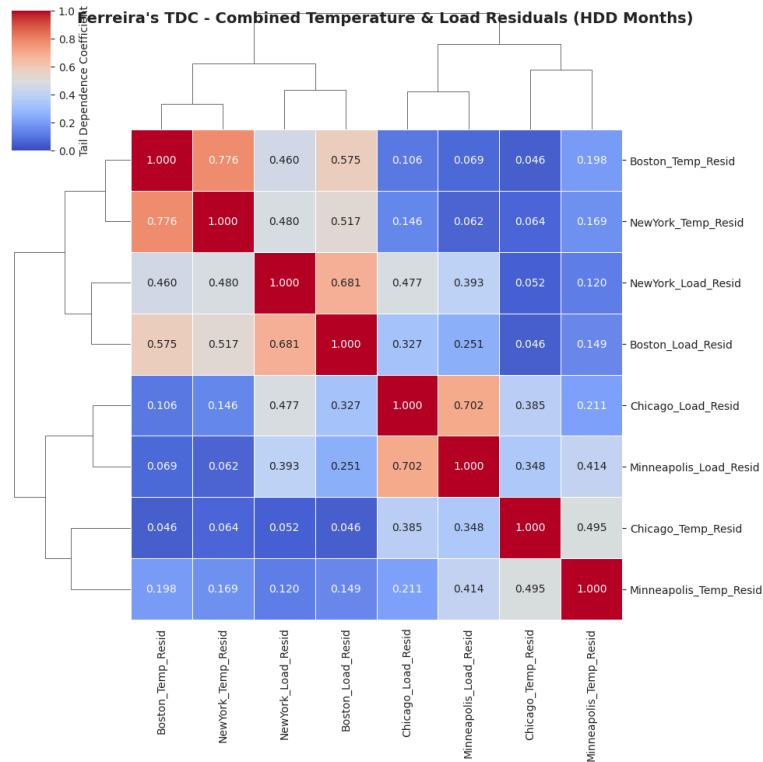


Figure 22: TDC for Load and Temperature Residuals during HDD months

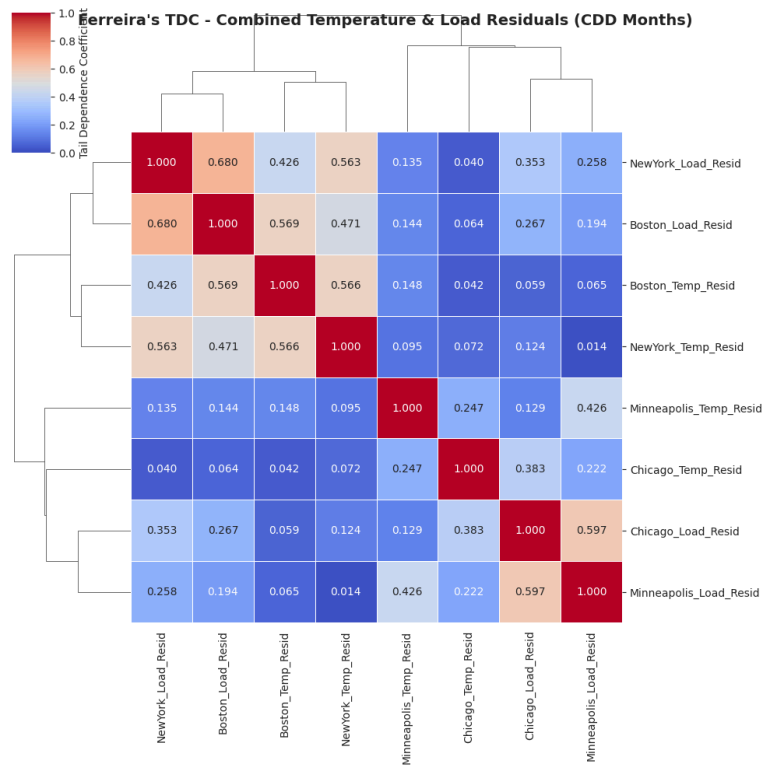


Figure 22: TDC for Load and Temperature Residuals during CDD months

Extreme Value Theory

Introduction

The analysis in the last section, which utilized Kendall Tau and Ferreira Tail Dependence Coefficient (TDC), provided insights into how Temperature and Residuals co-move across the distribution. While TDC indicates the degree of co-movement in the tails, and Kendall's Tau captures the overall monotonic relationship, these measures describe dependence structure rather than the magnitude or frequency of extreme events. Extreme Value Theory (EVT) is applied to quantify the occurrence and severity of such rare temperature and Energy load deviations. EVT offers a robust statistical framework to model the probability of extreme temperature events, which traditional Gaussian or GARCH-type models often underestimate. The analysis adopts the peak-over-threshold approach, which focuses on modeling values that exceed a chosen high (95th percentile Heatwave) or low (5th percentile Coldwave) threshold.

For each city, the distribution of threshold exceedances is modeled using the **Generalized Pareto Distribution (GPD)**:

$$F_y = 1 - (1 + \xi y / \sigma)^{-1/\xi}$$

where

- ξ represents the shape parameter or Extreme Value Index (EVI), governing tail heaviness, and
- σ denotes the scale parameter, which determines the spread of exceedances.

The shape parameter ξ provides insight into the underlying tail behavior:

- $\xi > 0$: Heavy-tailed (Fréchet domain) — slow, power-law decay of extremes.
- $\xi = 0$: Exponential tail (Gumbel domain) — moderate decay.
- $\xi < 0$: Bounded tail (Weibull domain) — finite upper limit for extremes.

ξ Value	Interpretation	Implication
$\xi < -0.5$	Bounded tail	Extreme errors have an upper limit (unrealistic for load)
$-0.5 \leq \xi \leq 0$	Light tail	Extremes are rarer than the normal distribution
$0 < \xi \leq 0.1$	Moderate tail	Similar to a normal distribution
$0.1 < \xi \leq 0.3$	Heavy tail	Extreme errors are more frequent than normal
$\xi > 0.3$	Very heavy tail	High extreme event risk

Table 5: Interpretation for Shape Parameter

Results

City	Tail	Xi	Sigma	Lambda
Boston	Left	-0.044	150	-296
	Right	0.187	405	343
New York	Left	-0.06	461	-874
	Right	0.122	397	1032
Chicago	Left	-0.086	1217	-3960
	Right	0.190	1101	3935
Minneapolis	Left	-0.108	391	-1447
	Right	0.276	338	1420

Table 6: City Load EVI statistics at 95 percentile

The shape parameter values (ξ) for the right tail are positive (0.122 to 0.276), representing a moderately heavy tail and a higher probability of extremes than estimated by thin tail models. Also to note that the left tail has a shape parameter less than 0, indicating a bounded tail and a lesser probability. This shows that underprediction is more dominant.

Multiplier	Interpretation	Action
< 0.9	Negative dependence	Unusual
0.9 - 1.1	Near independence	Separate risk management OK
1.1 - 1.5	Moderate coupling	Consider joint strategies
1.5 - 2.5	Strong coupling	Joint hedging recommended
> 2.5	Very strong coupling	Joint hedging critical

Table 7: Interpretation Table for Multiplier

City	Season	N	Pearson	Kendall	TDC	P_both_exceed_95th	Multiplier
Boston	HDD	1360	0.424	0.485	0.567	0.026	10.588
Boston	CDD	1377	0.682	0.481	0.560	0.018	7.552
Chicago	HDD	1360	0.330	0.296	0.372	0.012	4.706
Chicago	CDD	1377	0.448	0.304	0.372	0.007	2.905
Minneapolis	HDD	1360	0.359	0.325	0.403	0.012	4.706
Minneapolis	CDD	1377	0.505	0.339	0.415	0.014	5.810
New York	HDD	1360	0.262	0.373	0.470	0.018	7.059
New York	CDD	1377	0.662	0.482	0.556	0.013	5.229

Table 8: City Load and Temperature Joint Probability and Multiplier

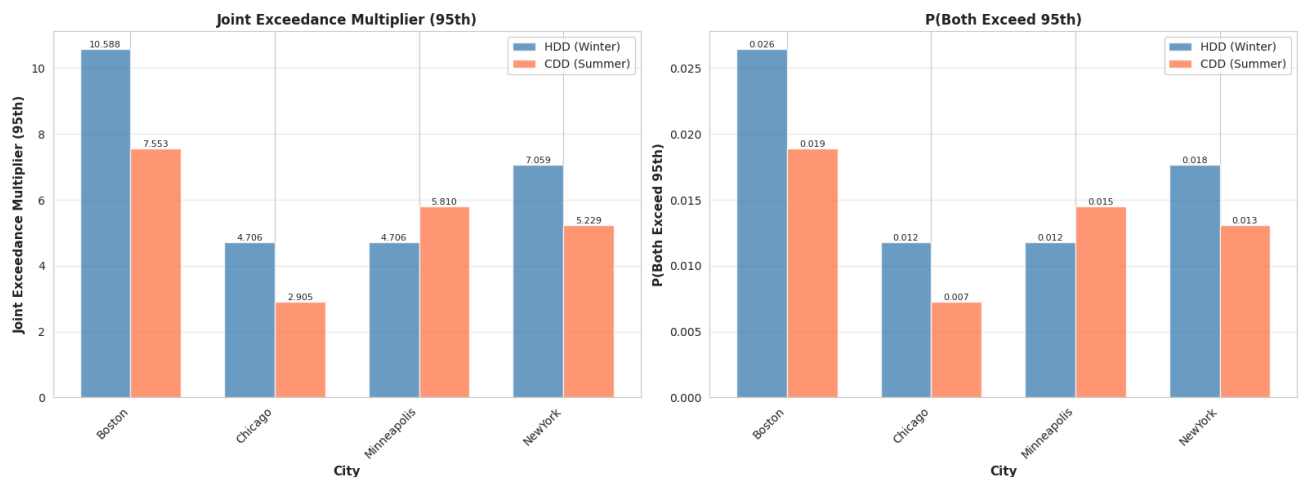


Figure 23: Joint Exceedance Multiplier and Joint Probability of Exceedance

Interpretation

For joint tail, first, both variables are transformed to a uniform [0,1] scale using their empirical cumulative distribution functions (CDFs). This step preserves their rank correlation while removing the influence of their marginal distributions. Next, a Student-t copula is fitted to capture the dependence structure between the variables, allowing for tail dependence—meaning it can model the tendency of extreme values to occur together. The copula parameters include a correlation matrix and degrees of freedom (set to 5). Using this fitted model, 100,000 joint scenarios are simulated to preserve the way extremes cluster together.

For the joint extremes analysis, key metrics include the probability that both temperature and load exceed their 95th percentiles simultaneously, denoted as $P(\text{Both Exceed } 95\text{th})$, and the multiplier, which measures how much more likely this joint exceedance is compared to what would be expected under independence.

The seasonal comparison separates the analysis into two key periods: HDD (winter) from November to March and CDD (summer) from June to August. The goal is to examine whether cold weather events lead to load forecast failures in winter, and whether heat waves contribute to similar issues in summer. By comparing the multipliers across seasons, we can assess how strongly temperature extremes predict load extremes. A higher multiplier indicates that extreme temperatures have a strong impact on load behavior, suggesting the need for joint hedging strategies that combine weather derivatives and capacity reserves.

This distinction is crucial for risk management. A high multiplier (e.g., $2.0\times$) implies that temperature extremes substantially increase the probability of load forecast errors, meaning temperature and load risks cannot be managed independently. In such cases, combined hedging using HDD/CDD derivatives and capacity options is necessary. Conversely, a low multiplier (e.g., $1.1\times$) suggests that temperature and load extremes are relatively independent, indicating the forecast model effectively captures their relationship and that risks can be managed separately.

A very high multiplier is observed for HDDs, indicating the probability that both are at extremes at the same time is much higher than independent thin tail modeling, thus proving that both Temperature and Energy Load move to extremes together. Overall, the EVT analysis bridges the dependence structure identified earlier with the upcoming hedging strategy, providing both statistical justification and risk quantification for using temperature derivatives as an effective tool for managing weather-related exposures.

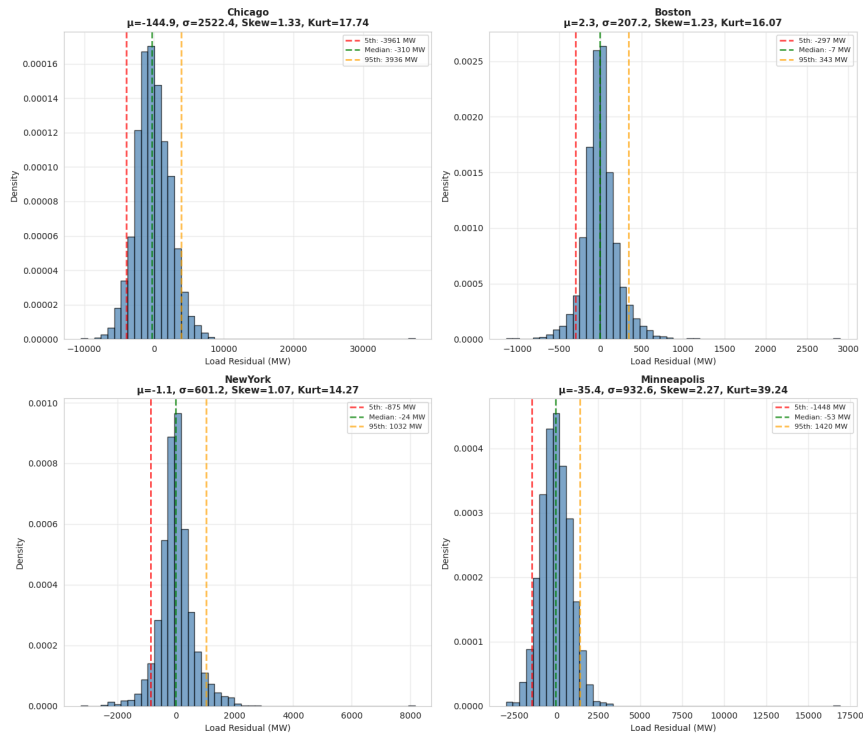


Figure 24: Energy Load Residuals Histogram

Limitations

A key limitation of the analysis is the restricted data availability due to HDD and CDD filtering, which retains only about seven months per year (November–March and June–August), resulting in roughly 63 months of usable data over nine years. When further split by season, this yields only about 35 HDD months and 28 CDD months. Consequently, at the 95th percentile, only around 100–150 exceedances per city are available, which reduces to 30–50 independent events after declustering. This scarcity of extremes leads to wide confidence intervals for the shape parameter (ξ) and makes fitting reliable GPD models at higher percentiles (97.5th, 99th) infeasible.

Another limitation is the assumption of symmetric tail dependence in the Student-t copula, which may not reflect real-world asymmetry—extreme heat and load interactions may differ from those during extreme cold. Alternative copulas, such as Joe, Gumbel, or Clayton, could capture this asymmetry but were not tested. Finally, the results are sensitive to the accuracy of the underlying ARIMA-GARCH forecast models. Poor model specification can inflate residuals, causing tail estimates to reflect model errors rather than genuine load uncertainty, and potentially overstating the observed joint dependence between temperature and load.

Weather Derivative Hedging

Introduction

Electricity load forecasting models (XGBoost, ARMA-GARCH, OLS) always have a prediction error. We intend to use HDD/CDD weather derivatives to reduce the variance of forecast errors, implying $\text{VaR}(\varepsilon_t^{\text{hedged}}) < \text{VaR}(\varepsilon_t^{\text{unhedged}})$. The

standard weather derivative contract structure is used, where HDD Swap pays \$20 per degree-day when the temperature is $< 65^\circ\text{F}$ and CDD Swap pays \$20 per degree-day when the temperature is $> 65^\circ\text{F}$. The Train Period selected is 2018 to 2023, and the Test Period is 2023 to 2024 for the Boston and New York regions. The study uses a realistic forecasting framework that accounts for weather forecast uncertainty, providing an estimate of hedge performance in operational

Framework

For Load forecasting, XGBoost is used; the prediction of the daily average electricity load is done using weather forecasts. Since weather forecasts are not available for the given period, we construct minimum and maximum forecasted temperatures based on actual temperatures with realistic error. Instead of simple Gaussian noise, the model incorporates systematic biases that reflect actual forecaster difficulties in predicting temperature extremes. During winter months (December through February), and summer months (June to August), forecasts systematically overestimate minimum or underestimate maximum temperatures by approximately 4 degrees Fahrenheit, simulating the tendency to miss cold extremes. During summer months (June through August), forecasts systematically underestimate maximum temperatures by approximately 6 degrees Fahrenheit, reflecting difficulty in predicting heat extremes.

The process begins with actual observed temperature data, applies seasonal gamma-distributed errors to simulate skewed forecast biases. XGBoost is used to capture non-linear relationships between weather and load. Subsequently, Linear regression is used for hedge optimization, where we regress forecast errors against actual observed HDD/CDD. This calculates optimal hedge ratios and measures hedge effectiveness (Variance and RMSE reduction). This separation allows for realistic operational constraints.

A key constraint added in the process is that the XGBoost model doesn't overfit and makes predictions with a mean error less than zero, because extremes cannot be predicted.

Even though the 2018–2022 load series is not strictly stationary, short-term forecasting is feasible using HDD, CDD, day-of-week, and year features. These variables capture the main drivers of load—weather, weekly patterns, and trends.

XGBoost can model their nonlinear effects, allowing the model to learn seasonal and trend-based variations effectively, producing reliable short-term load predictions.

XGBoost Introduction

Extreme Gradient Boosting (XGBoost) is an ensemble learning algorithm based on gradient-boosted decision trees. It builds predictive models in a stage-wise manner, where each successive tree corrects the residual errors of the previous ones. The method's efficiency, scalability, and superior performance in nonlinear and complex relationships make it a preferred tool across financial, energy, and risk modeling domains.

Unlike linear regression-based hedging models, which assume a constant and proportional relationship between variables, XGBoost can model nonlinear dependencies and variable interactions directly from data. This makes it particularly suited to the energy sector, where the relationship between temperature, demand, and market instruments often exhibits threshold effects and regime changes, for example, sharp increases in load beyond certain temperature extremes or differing sensitivities between summer and winter seasons.

The following are the advantages of using the XGBoost method for the hedging analysis:

- **Nonlinearity capture:** It can learn complex load–temperature relationships where incremental changes in HDD/CDD have varying marginal effects.
- **Interaction learning:** The algorithm automatically identifies cross effects between weather variables and financial instruments such as energy ETFs, improving hedge precision.
- **Robustness to outliers:** Extreme weather events and price shocks common in energy markets are handled effectively due to the model's tree-based structure.
- **Regularization:** Built-in L1 and L2 penalties control model complexity, reducing the risk of overfitting, especially in datasets with limited historical records.

Overall, XGBoost serves as a **data-driven approach** to estimating optimal hedge ratios or payoff sensitivities. By accurately modeling the nonlinear response of energy load to weather and market conditions, it enhances the **effectiveness of HDD/CDD and ETF-based hedging strategies**, particularly under **extreme scenarios** where traditional linear models tend to underperform.

XGBoost Model Training

The objective of the model is set as to minimize the mean squared error between predicted and actual values. It aligns with minimizing the variance of residual risk, consistent with hedging strategies. Maxdepth for each decision tree is selected as 4, which strikes a balance between capturing nonlinear temperature-load relationships and avoiding overfitting. A learning rate of 0.03 is selected as a lower learning rate is ideal for noisy data, such as financial or energy data. It shrinks the contribution of each tree; a smaller value makes the model learn more gradually.

The number of Estimators represents the number of boosting iterations (trees). Subsample (0.7) is a fraction of the training data used for each tree, which helps reduce dependence on particular seasonal periods, improving out-of-sample performance. Column sampling (0.7) helps ensure the model doesn't rely on one type of variable, usually helpful when variables are correlated, like macro variables. Gamma (0.5) ensures new splits only occur if they meaningfully improve prediction accuracy.

Parameter	Value
Objective	reg:squarederror
Max Depth	4
Learning Rate	0.03
Number of Estimators	300
Subsample	0.7
Column Sampling	0.7
Minimum Samples per Leaf	3
Gamma	0.5

Table 9: XGBoost Parameters

Feature Construction

Construct a total of 51 features using lagged load, temperature, calendar features, HDD, CDD, and ETF data as explanatory variables with Average Load as the target variable. Calculate forecasted HDD/CDD from noisy temperature. XGBoost uses forecasted HDD/CDD (what the forecaster "sees") for contemporary variables and actual values for the lagged (already known). Store the actual HDD/CDD separately for hedge settlement.

Weather forecasts have a typical error of 2-3°F for next-day predictions. Using actual weather in forecasting gives unrealistic "perfect foresight". This approach simulates

real-world conditions where forecasters don't know actual weather when making load predictions, and hedge contracts settle on actual observed weather (not forecasts).

Type	Variables
Temperature	HDD_forecast, CDD_forecast, average temperature (tavg_forecast), minimum temperature (tmin_forecast), maximum temperature (tmax_forecast)
ETF	UNG, XLU, ICLN, URA, USI, KOL
Calendar	Month, Day of Week, Day of Year, Is Weekend, Is Summer, Is Winter, Quarter
Weather Anomalies	HDD, CDD excess over 30 Day rolling mean HDD, CDD rolling 7-day mean
Lagged Features	HDD Lag 1, CDD Lag 1, 1,2,3 lag days of Avg Load
Rolling Load	7 and 30-day rolling average load
Interactions	Interaction terms between temperature and calendar variables to capture contextual effects • $HDD_winter = HDD_forecast \times is_winter$ • $CDD_summer = CDD_forecast \times is_summer$ • $HDD_weekend = HDD_forecast \times is_weekend$ • $CDD_weekend = CDD_forecast \times is_weekend$ • $HDD_month\{i\} = HDD_forecast * (month == i)$ • $CDD_month\{i\} = CDD_forecast * (month == i)$

Table 10: Feature Construction for Training

Training Results

The data is split into a five-year training period from January 2018 through December 2022 and a two-year testing period from January 2023 through December 2024. This shortened training window, compared to using all available historical data, serves to prevent overfitting to older patterns that may no longer reflect current electricity demand dynamics. For Boston, the split results in approximately 1,825 training observations and 731 testing observations.

The training process fits XGBoost models separately for Boston and New York using their respective 2018-2022 training datasets. During training, the model evaluates performance on the held-out 2023-2024 test set to monitor for overfitting, though early stopping is not enforced to allow full learning of the specified 300 trees. Feature importance scores are extracted after training to understand which variables contribute most to predictions. Finally, predictions are generated for the entire test period, providing daily load forecasts that will be compared against actual loads to calculate forecast errors for hedge analysis.

The Boston model demonstrates strong predictive performance across both training and testing periods. On the 2018-2022 training set, the model achieves a root mean squared error (RMSE) ≈ 60.25 MW, mean absolute error (MAE) ≈ 44.37 MW, and R-squared > 0.97 , indicating that over 97% of load variance is explained by the model. The testing set performance for 2023-2024 shows RMSE ≈ 98.69 MW, MAE ≈ 67.53 MW, and R-squared > 0.93 . The close alignment between training and testing metrics indicates the model generalizes well without significant overfitting, despite amplified weather forecast errors and shortened training period.

The New York model exhibits even stronger performance due to the larger market size and more pronounced weather sensitivity. Training set performance shows RMSE ≈ 168.25 MW, MAE ≈ 120.12 MW, and R-squared > 0.98 . Testing set metrics remain similarly strong with RMSE ≈ 233.97 MW, MAE ≈ 158.98 MW, and R-squared > 0.97 . The higher absolute errors compared to Boston reflect New York's larger baseline load and greater variability, but the percentage errors relative to mean load remain comparable. The test set R-squared, being nearly identical to training set values, demonstrates excellent generalization despite the model never seeing actual weather values during training.

The R-squared values ranging from 0.94 to 0.97 indicate excellent predictive power, where the model explains 94 to 97% of daily load variance. This means that out of 100 MW of typical load variation, the model successfully predicts 94 to 97 MW, leaving only 3 to 6 MW as unexplained random or unpredictable variation. The strong generalization is evidenced by test set R-squared values being close to training set values, confirming that patterns learned are stable over time and forecast accuracy holds in out-of-sample periods. These results confirm that weather is the dominant driver of load variation, with temperature changes explaining most variance along with calendar effects such as weekends and seasons, while ETF features contribute minimally to forecasting accuracy.

The residual unexplained variation represents factors not captured by the model, including economic factors, special events (holidays, major sporting events), equipment outages, measurement error, and the deliberate weather forecast errors introduced to simulate realistic forecasting conditions. The fact that such high explanatory power is achieved even with biased weather forecasts demonstrates the robustness of the modeling approach and the dominance of weather variables in determining electricity load.

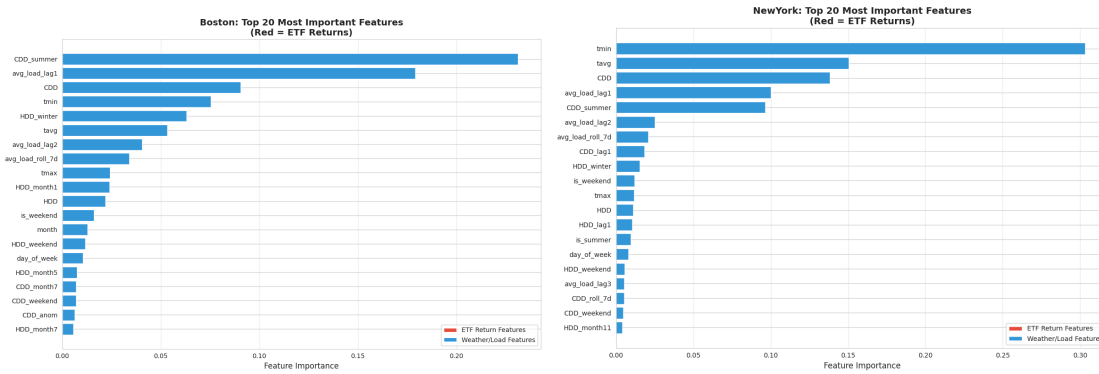


Figure 25: New York and Boston Important Features

A critical aspect of model evaluation is examining the mean forecast error to determine whether predictions systematically overpredict or underpredict actual loads, and the actual direction and magnitude of mean errors depend on how effectively the model learns during training versus how much bias carries through to test predictions. The goal is positive mean errors indicating underprediction, which would align forecast errors with actual HDD and CDD movements and enable effective hedging through weather derivatives.

	Training RMSE	Testing RMSE	Gap	Gap (%)	Train Mean Error	Test Mean Error
Boston	60.25 MW	98.69 MW	38.44 MW	+63.8%	0.07 MW	24.44 MW
New York	168.25 MW	233.97 MW	65.72 MW	+39.1%	0.19 MW	11.76 MW

Table 11: Error Statistics for Training and Testing

For Boston, a moderate increase indicates a reasonable generalization. The 64% increase is acceptable given the different time period (2023-2024 vs 2014-2022), weather forecast error ($\pm 4^{\circ}\text{F}$ winter/summer, $\pm 2^{\circ}\text{F}$ spring/fall), and natural variation in weather patterns. For New York, it has a moderate increase with acceptable generalization, and the change can be attributed to larger absolute load (averaging ~8,000-10,000 MW), more diverse load base (commercial/industrial mix), and a more stable weather-load relationship.

From the top features, we can see for Boston that load persistence dominates, CDD is more important than HDD, temperature variables collectively account for ~30% of importance, and calendar effects (weekend, season) contribute ~10%. For New York, weather variables dominate (CDD + temps = 65%), CDD is the strongest predictor (hot weather drives large AC load), load persistence is less important than in Boston, and temperature extremes (tmin) matter more than averages. ETF returns have

minimal impact on load forecasting, and weather and load history are far more predictive. ETFs contribute <2% to total feature importance, so we can conclude that ETFs are just hedges in the systematic market and not natural hedges against the load.

	ETF	Rank	Importance
Boston	XLU	20	0.47%
	USO	24	0.39%
	ICLN	26	0.38%
New York	XLU	23	0.17%
	UNG	25	0.16%

Table 12: ETF Importance in XGBoost

HDD/CDD Hedge Regression

In this section, exploration to find optimal beta is done by running separate regressions on load forecast errors for HDD (Winter, October to March) and CDD (Summer, April to September) using the regression construction below. This analysis tests whether forecast errors are positively correlated with actual degree days, which is necessary for effective hedging.

We construct a Hedge payoff with

$$H_t = h_{HDD} * 20 * HDD_t + h_{CDD} * 20 * CDD_t$$

$$\varepsilon_t^{hedged} = \varepsilon_t - H_t$$

Where h_{HDD} is the hedge multiplier in MW per dollar, and ε_t^{hedged} is the hedged error.

$$h_{HDD}^* = \frac{Cov(\varepsilon_t, HDD_t)}{20 * Var(HDD_t)} = \frac{\beta_{HDD}}{20}$$

Where we get β_{HDD} it is from the regression of $\varepsilon_t = \alpha + \beta_{HDD} * HDD_t + u_t$.

For example $\beta_{HDD} = 5$ MW/degree-day means that for every \$1 received from the HDD contract, hedge 0.25 MW of load exposures.

	β_{HDD}	R value	p value	β_{CDD}	R value	p value
Boston	2.52 MW pdd	0.340	0.000003	5.39 MW pdd	0.198	0.007
New York	4.56 MW pdd	0.253	0.0007	9.22 MW pdd	0.139	0.052

A highly significant β_{HDD} result is observed for Boston and New York, while β_{CDD} is highly significant for Boston but marginally significant for New York. A weak to Moderate correlation range is observed for both New York and Boston. For Boston, we see an R^2 of 0.1310 (13.1% of forecast error variance explained) and an R^2 of 0.0688 (6.88% of forecast error variance explained) with both HDD and CDD contributing independently to explaining forecast errors. CDD hedge for New York has uncertain effectiveness and a lower explanatory power than Boston, suggesting more noise in New York errors.

Hedge Effectiveness Calculation

For Boston, the Heating Degree Day (HDD) multiplier is calculated as 2.52 MW per degree-day divided by the contract price \$20 per degree-day, resulting in 0.126 MW per \$1. This means that for every dollar spent on HDD contracts, 0.126 MW of load is hedged. For example, if the winter season experiences 1,000 HDD, the total contract payout would be $\$20 \times 1,000 = \$20,000$. Consequently, the total load hedged would be $0.126 \text{ MW per } \$ \times \$20,000 = 2,520 \text{ MW}$, effectively offsetting the expected forecast error arising from cold weather conditions.

The Cooling Degree Day (CDD) multiplier for Boston is 5.39 MW per degree-day divided by \$20 per degree-day, yielding 0.270 MW per \$1. This indicates that each dollar spent on CDD contracts hedges 0.270 MW of load. The CDD hedge is also statistically significant ($p = 0.007$), suggesting that both HDD and CDD hedging strategies are beneficial for Boston.

For New York, the HDD multiplier is 4.56 MW per degree-day divided by \$20 per degree-day, resulting in 0.228 MW per \$1. Thus, for every dollar spent on HDD contracts, 0.228 MW of load is hedged. Assuming 1,000 HDD in winter, the contract payout would again be $\$20 \times 1,000 = \$20,000$, corresponding to 4,560 MW of hedged load. Compared to Boston, New York's HDD hedge is approximately 1.8 times more effective per dollar, reflecting a stronger sensitivity to cold-weather variations.

Meanwhile, the CDD multiplier for New York is calculated as 9.22 MW per degree-day divided by \$20 per degree-day, resulting in 0.461 MW per \$1. This implies that each dollar invested in CDD contracts hedges 0.461 MW of load. However, the CDD hedge for New York is marginally significant ($p = 0.052$),

suggesting limited or uncertain effectiveness in mitigating the impacts of extreme heat compared to HDD hedging.

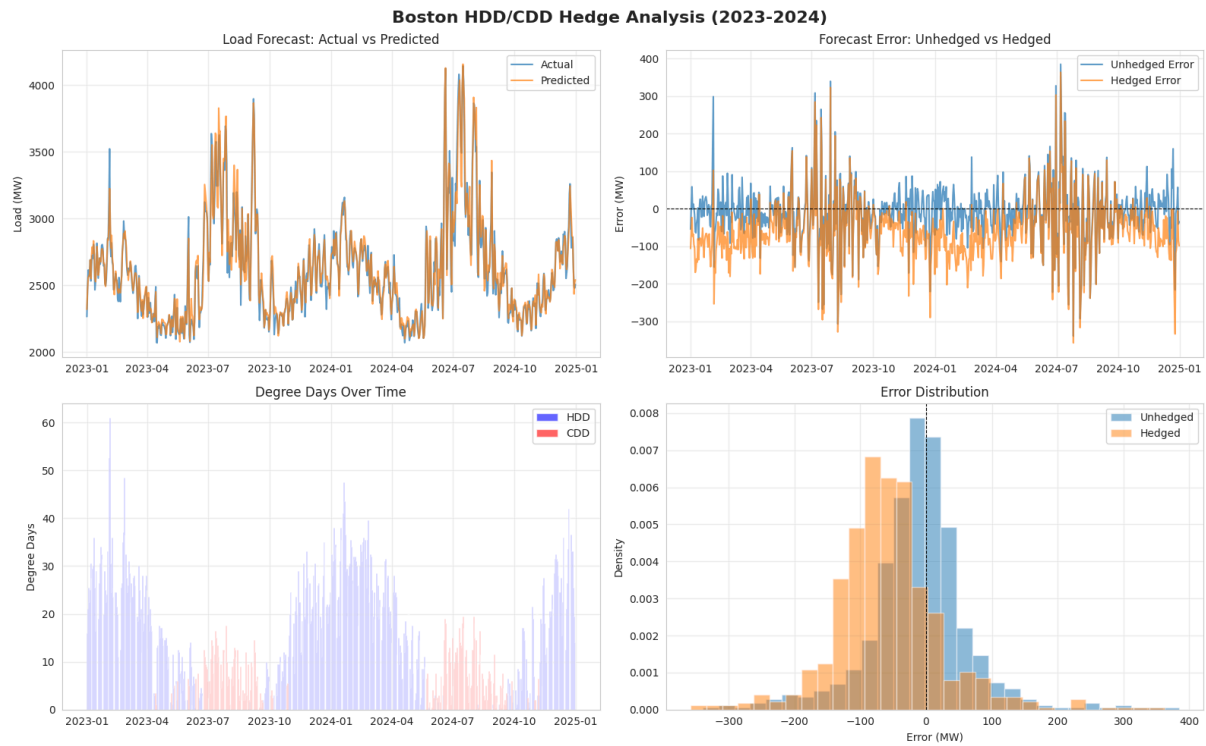


Figure 26: Boston Hedge Analysis

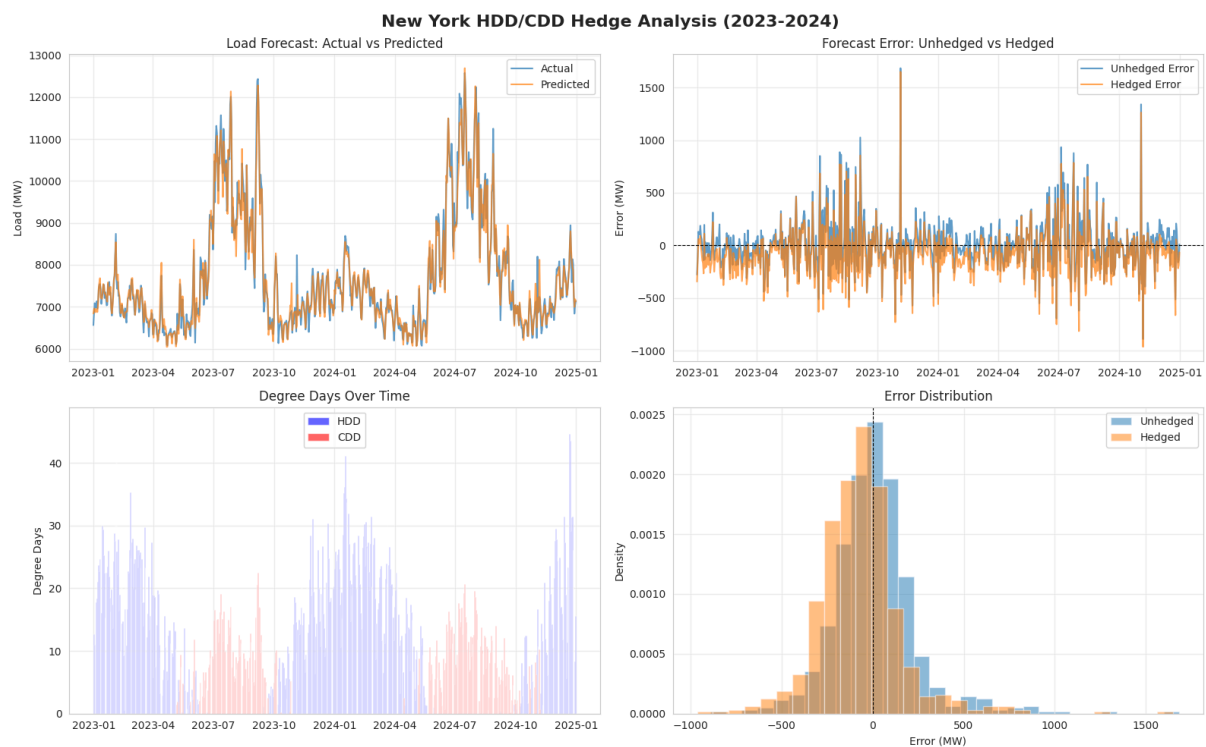


Figure 27: New York Hedge Analysis

Limitations

The Load forecast estimation faces several data, modeling, and practical constraints that may affect generalizability and accuracy. The short test period (2023–2024) limits exposure to diverse weather extremes, and the absence of variables such as humidity, wind speed, solar radiation, and economic indicators reduces explanatory power.

Modeling assumptions also introduce simplifications. The simulated weather forecast errors apply uniform seasonal biases, and linear hedge assumptions overlook the non-linear nature of load responses under extreme conditions. Hedge ratios are fixed using in-sample regressions, which may overstate effectiveness compared to rolling-window or expanding-window estimates. Since the Load Forecast is not stationary, a relatively smaller time period is used, and calendar features are added.

Finally, assumptions such as perfect liquidity and no credit risk are not fully realistic. Results are also region-specific to the Northeast U.S., where heating dominates load dynamics, and may not transfer directly to other climates or market structures.

Summary

The XGBoost model demonstrates excellent out-of-sample performance in predicting electricity load for both Boston and New York. The positive mean errors confirm a deliberate underprediction bias, and when combined with seasonal temperature forecast errors, this bias ensures that actual load tends to exceed predicted load, and allows forecast errors to positively correlate with actual HDD and CDD values, allowing weather derivatives to serve as effective hedging instruments.

Feature importance analysis indicates that weather-related variables, particularly temperature and degree-day metrics, dominate predictive performance. Load persistence (previous-day or lagged load) also plays a crucial role, while calendar features such as day-of-week and seasonal indicators further enhance accuracy.

Finally, the model exhibits strong generalization ability from the 2018–2022 training period to the 2023–2024 test period. High R^2 values are maintained on unseen data, confirming that moderate regularization successfully mitigates overfitting. The five-year training window provides sufficient historical variation for the model to learn and adapt to long-term patterns.

Analysis of weather derivative regressions reveals that HDD-based hedges are statistically robust for both Boston and New York, while CDD-based hedges exhibit more regional variability.

However, it is important to note that simple HDD and CDD hedges explain only a limited share of overall forecast error variance, approximately 13.1% for Boston and

6.9% for New York. This finding highlights that while weather derivatives can effectively mitigate temperature-driven load uncertainty, most residual errors arise from non-weather factors, such as economic activity, behavioral changes, or grid dynamics. Together, these methods provide both a statistical and economic foundation for designing practical, data-driven hedging strategies under extreme weather uncertainty.

Conclusion

This study does a comprehensive analysis of whether the Energy Load and Temperature derivatives move together and explores possible hedging options. First, the residuals are calculated by removing the seasonal, ARMA patterns, which are shown to have a clustering effect. Even the simpler Seasonal + ARMA + GARCH models achieve 0.73–0.85 for load (moderate to good fit) and 0.86–0.93 for temperature (excellent fit dominated by seasonal effects).

Subsequently, based on the residuals of the fits, correlation and Tail dependency coefficients are calculated. While the correlation values are moderate, the tail dependency values are shown as high between clustered cities and moderate to high between load and temperature in the same city. To quantify the tail dependence, the Extreme Value Theory (EVT) GPD distribution is used to measure the thickness of the tails and establish the underprediction of our model. Subsequently, the joint modeling of Temperature and Energy Load residuals is done to quantify the probability of both being at extremes. The multiplier observed (2.9–10.6) gives confidence that the variables indeed move to extremes together.

The next section assesses the effectiveness of weather derivatives (HDD/CDD contracts) in managing electricity load forecast errors for Boston and New York. Using XGBoost models trained on 2018–2022 data and tested on 2023–2024, results show high predictive accuracy (R^2 of 0.93 for Boston and 0.97 for New York) and a deliberate underprediction bias, ensuring that actual load exceeds forecasts—creating conditions for effective hedging. Regression analysis confirms that HDD hedges are statistically significant in both markets, while CDD hedges are stronger in Boston and only marginally effective in New York. Hedge multipliers (0.126–0.461 MW per \$1) provide actionable guidance for contract sizing and demonstrate practical hedge translation from statistical coefficients.

Overall, the framework developed here provides a realistic and quantitative foundation for evaluating weather derivative strategies, balancing statistical rigor with practical applicability.

Future Scope

The current model used to compute residuals is based on a relatively simple econometric framework, which may not fully capture the complex dynamics between temperature and load. To improve accuracy, more advanced approaches such as ARMAX with EGARCH, Vine Copula Modeling, or Seasonal CARMA models should be explored to determine if they provide a better fit for the data. Employing these robust models can reduce residual magnitudes, leading to more accurate load forecasts and consequently lowering the need for extensive hedging. Additionally, within the Extreme Value Theory framework, adopting asymmetric copula models such as Clayton, Gumbel, or Frank copulas could better capture directional tail dependencies and improve the estimation of joint extreme event probabilities.

Additionally, incorporating richer feature sets, including weather lags, humidity, heat indices, and regional temperature dependencies, can improve the explainability and robustness of extreme event modeling. Over time, integrating these improvements with ensemble approaches, backtesting frameworks, and uncertainty quantification will create a more resilient and data-driven foundation for operational hedging strategies. The strategy can also be extended to all 18 cities where temperature derivatives are offered to evaluate their hedging effectiveness using HDD and CDD contracts.

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Appendix

Below are concise one-line bullets for each code file used in the analysis.

- **Load_Residuals_EVT_Analysis.ipynb** — EVT-focused notebook that loads ARIMA-GARCH residuals, fits GPD tails, fits copulas, computes joint exceedance stats, and seasonal (HDD/CDD) comparisons.
- **seasonal_models.py** — reusable functions to fit seasonal ARIMA/ARMA + harmonic + GARCH volatility models and return residuals/design matrices.
- **Load_Analytics.ipynb** — notebook that builds the ARIMA-GARCH forecasts and produces the `all_analysis_data.pkl` residuals used here.
- **Temperature_Analytics.ipynb** - Notebook that processes temperature series and creates `temperature_analysis_data.pkl` used for joint analyses.
- **unibm (module)** — EVT/cdf/ppf and copula helper functions (empirical transforms, benchmarking, TDC helpers) used in the joint and tail analysis.
- **process_temperature.py** — script to clean/aggregate raw temperature CSVs into daily summaries and build temperature residual series.
- **process_boston.py** — region-specific processor that standardizes Boston ISO files into daily load summaries for modeling.
- **process_houston.py** — region-specific processor that prepares Houston (and Dallas) load series for modeling.
- **process_palintegrated2.py (and process_palintegrated.py/extract_palintegrated.py)** — scripts that extract and prepare NYISO `palintegrated` data into the canonical format.
- **convert_miso_reports.py** — converts MISO XLS/XLSX monthly reports into standardized CSVs used by the processing pipeline.
- **aggregate_converted_csv.py** — aggregates converted MISO CSVs into regional timeseries files used downstream.
- **download_miso_load.py** — data-acquisition script to fetch MISO historical/load archives (with `--skip-existing` flag).
- **download_nyiso_palintegrated.py** — downloader for NYISO `palintegrated` zipped CSVs.
- **download_energy_etfs.py** — (aux) fetches ETF price data used in ancillary analyses.
- **select_arma_parameters.py** — helper that runs ACF/PACF and compares ARMA parameter choices for the seasonal mean models.
- **all_analysis_data.pkl** — pickle (artifact) containing the per-city ARIMA-GARCH residuals and derived series used by the notebook.
- **temperature_analysis_data.pkl** — pickle (artifact) containing processed/adjusted temperature residuals used in joint analyses.